# Questions of the day

- How can we compare algorithms without ever implementing them?
- What does $O(n^2)$ *really* mean?

# Big-Oh notation

**William Hendrix**

*Lecture 1*

# Today

- Syllabus
- RAM model of computation
- Big-Oh notation
  - Motivation
  - Proof review
  - Formal definitions

# Syllabus

- Read at: sit.instructure.com
- Contact: whendrix@stevens.edu
- Office hours:
  - MT 2:00-3:00 pm
  - GS 251
- CAs: Milan Chandiramani and Gangqun (David) Yang
- Office hours:
  - Milan: Tuesdays 5:30-6:30pm in GS 347 and
    Wednesdays 5-6pm in GS 224
  - David: Fridays 3-5pm in GS 433

- Course objectives, grading scale, exams, etc.
- Class participation
- Feedback form
- Slides

# What will we learn in this class?

- **How to determine if an algorithm is efficient**
  - RAM model
  - Big-Oh definition and properties
- **How to improve your algorithms by organizing data**
  - Stacks and queues
  - Binary search trees
  - Priority queues and heaps
  - Hash tables
- **How to develop your own algorithms**
  - Greedy algorithms
  - Divide-and-conquer
  - Dynamic programming
  - Graph traversals
  - Backtracking
- **Classical sort, search, and graph algorithms**

# Brainstorm: time complexity

- What are the factors that contribute to the running time of an algorithm?
  - Processor speed
  - Number of instructions executed
  - Cache coherency
  - Resource conflicts (network, hard disk, etc.)

- Which of these are important when comparing algorithms?
  - Processor speed affects fast and slow algorithms equally
    - Not an important factor

- What can we *most reliably* control when designing an algorithm?
  - Number of instructions executed

# RAM model of computation

- Set of assumptions that make analysis more reasonable

**<u>Assumptions</u>**

1. All "basic" operations (assignment, arithmetic, branching, memory access, etc.) take 1 operation
   - Loops and functions do not qualify
2. We have "infinite" memory

**Cons**

- Different operations take different number of clock cycles
  - Cache locality has significant impact
- Virtual memory can slow performance

**Pros**

- Can actually analyze algorithms

# RAM model example

**Input:** $data$: array of integers
**Input:** $n$: size of $data$
**Output:** index $min$ such that
$data[min] \leq data[j]$, for all $j$
from 1 to $n$

```
Algorithm: FindMin
min = 1
for i = 2 to n do
    if data[i] < data[min] then
        min = i
    end
end
return min
```

# RAM model example

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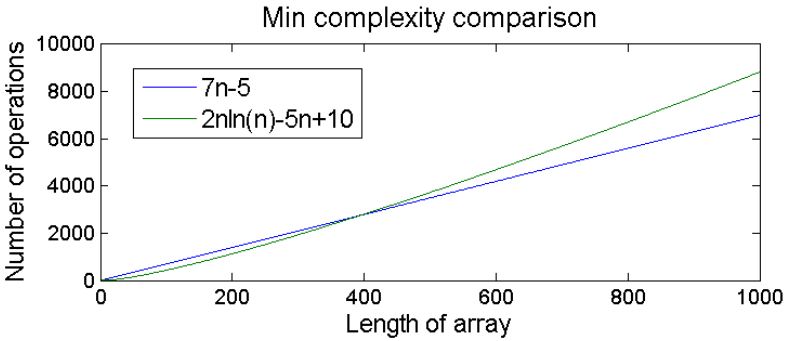
| Ops per line | Times executed |
|---|---|
| 1 | 1 |
| 2 | n-1 (arguably n) |
| 3 | n-1 |
| 1 | ??? (≤ n-1) |
| 0 | n/a |
| 1 | n-1 |
| 1 | 1 |

**Total ops:** ≤ 7(n-1) + 2 = 7n - 5

**Question:** is this better or worse than an algorithm that takes at most $2n \ln n - 5n + 10$ ops?

Better unless n < 395



# Big-Oh notation

- Technique for *abstracting away details* of complexity
  - Can be used for time complexity, space complexity, etc.

- **Main idea:** most important aspect of complexity is *how fast it grows* relative to input size
  - Focus on asymptotic (eventual) growth rate
  - "Fast" functions will eventually pass "slow" functions for large *n*
  - Coefficients only matter if growth rate is similar
  - Predicting behavior for small *n* is difficult and often pointless

- Big-Oh notation
  - Organizes growth rates into classes
  - Three main symbols: $O(f(n)), \Omega(f(n)), \Theta(f(n))$
    - Analogous to "at most", "at least", and "similar to" *f(n)*

# Justification of Big-Oh

- Algorithm runtime with $c$=1, running at 1 GHz:

| n= | lg($n$) | $n$ | $n$lg($n$) | $n^2$ | $n^3$ | $2^n$ | $n$! |
|---|---|---|---|---|---|---|---|
| 10 | 3 ns | 10 ns | 33 ns | 100 ns | 1 µs | 1 µs | 3.6 ms |
| 20 | 4 ns | 20 ns | 86 ns | 400 ns | 8 µs | 1 ms | 77 yrs |
| 30 | 5 ns | 30 ns | 147 ns | 900 ns | 27 µs | 1 s | |
| 40 | 5 ns | 40 ns | 213 ns | 1.6 µs | 64 µs | 18.3 min | |
| 50 | 6 ns | 50 ns | 282 ns | 2.5 µs | 125 µs | 13 days | |
| 100 | 7 ns | 100 ns | 644 ns | 10 µs | 1 ms | | |
| 1,000 | 10 ns | 1 µs | 9.97 µs | 1 ms | 1 s | | |
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| | | | | | | |
|---|---|---|---|---|---|---|
| "Fails" at: | Never! | billions | millions | 10k | 40ish | 16ish |

**Lesson:** on large data, coefficients not as important

# The formal definition

- We will use the *formal definition* of Big-Oh to *mathematically prove* claims about Big-Oh
- **<u>Goals</u>**
  - Understand exactly what Big-Oh means
  - Understand the difference between Big-Oh, Big-Theta, and Big-Omega
    - E.g., $O(n^2)$ vs. $\Theta(n^2)$ vs. $\Omega(n^2)$

- We'll start by reviewing mathematical proofs

# Mathematical proofs

- **Proof:** convincing mathematical argument for some *claim*
  - Each statement should follow what came before
  - May use theorems, definitions, algebra, etc.

- **Proof by example**
  - Proof technique for existential ("there is") claims
  - To show something exists, just give an example!

- **Example proof structure**
  - Consider *this thing*. Note that *it meets all the conditions of the problem for these reasons (…)*. Therefore, there must exist *something that meets these conditions*, namely *the thing from earlier*.
  - Adapt this structure to your problem

# Proof by example example

- Prove that 0 is even.
  - **Definition.** An integer $x$ is **even** if and only if there is some integer $k$ such that $x = 2k$.

- **Formal proof:**
  - Consider the integer $k = 0$.
  - Note that $2k = 2(0) = 0$.
  - Thus, there exists an integer $k$ such that $0 = 2k$, namely $k = 0$.
  - Therefore, 0 is even by the definition of even numbers.

- **Shorter version:**
  - 0 is even because $0 = 2(0)$.
- Still a valid proof <u>as long as</u> the reader can follow what you left out

# Direct proof

- Straightforward proof technique
  - Used to prove claims of the form "If A, then B"
- **Main idea:**
  - Assume "if" part is true
  - Argue that the "then" part is true

- **Example proof structure**
  - Suppose that *A is true*. *For these reasons (…)*, we can see that *B must be true*. Therefore, *if A, then B*.
  - May need several sentences for argument

# Direct proof example

- If $x$ is even and $x > 2$, $x$ must be composite.
  - **Definition.** An integer $x$ is **composite** iff there exist integers $a$ and $b$ greater than 1 such that $x = ab$.

- **Formal proof:**
  - Suppose that $x$ is even and $x > 2$.
  - Since x is *even*, there must be some integer $k$ such that $x = 2k$.
  - Since $x > 2$ and $x = 2k$, $k > 1$.
  - There must be two integers $a$ and $b > 1$ such that $x = ab$
    - Namely $a = 2$ and $b = k$.
  - Therefore, $x$ must be composite.

# Writing a proof

- Proofs are **not** created in the way they are written

1. Proof technique tells you how proof should begin and end
2. Apply relevant definitions
3. Manipulate things you know are true to move towards the goal
4. Work backwards from the goal towards the things you know
   - "How could I conclude this is true?"
5. Repeat 2-4 until the two sides meet
6. (*Optional*) Produce a formal write-up that starts at the beginning and continues to the end

- With practice, you can "see" the chain of logic more quickly

# Writing a proof

1. Use proof technique to write start and goal
2. Work forwards and backwards until solved

- **Example:** if $x$ is even and $x > 2$, $x$ is composite.
- Proof:
  - $x$ is even
  - $x > 2$
  - There exists an integer $k$ such that $x = 2k$
  - $2k > 2$
  - $k > 1$
  - $2 > 1$ and $k > 1$
  - There exists integers $a$ and $b > 1$ such that $x = ab$
    - Proof by example: find $a$ and $b$
    - *Hypothesis:* $a = 2$ and $b = k$
  - x is composite

What we know

Need to show

# Existential statements in proofs

- If we know something exists, we can give it a name
- **Example:**
  - There exists an even prime.
  - Let $x$ be that even prime number.
- **Important:** cannot be a variable used earlier in the proof

- **Invalid example:** prove that if $x$ and $y$ are even, $x = y$.
- **Proof:**
  - There exists some integer $k$ such that $x = 2k$,
  - and there exists some integer $k$ such that $y = 2k$.
  - Since $x = 2k$ and $y = 2k$, $x = y$.
- **Correct version:**
  - There exist integers $j$ and $k$ such that $x = 2j$ and $y = 2k$.
    - *Other options:* $k_1$ and $k_2$ or $k$ and $k'$
  - Since we don't know anything else about $j$ and $k$, we cannot conclude that $x = y$.

# Proof exercise

- Prove that if $x$ and $y$ are even, $x + y$ is even.
  - **Definition.** An integer $x$ is **even** if and only if there is some integer $k$ such that $x = 2k$.

If x and y are even, x = 2*k1 for some integer k1 and y = 2*k2 for some integer k2.
x + y = 2k1 + 2k2 = 2(k1 + k2)
Therefore, there exists an integer k3, k3 = k1 + k2, such that x + y = 2 *k3, so x + y is even.

# Proof exercise

- Prove that if $x$ and $y$ are even, $x + y$ is even.
  - **Definition.** An integer $x$ is **even** if and only if there is some integer $k$ such that $x = 2k$.

*Proof.* Since $x$ and $y$ are even, there must be integers $i$ and $j$ such that $x = 2i$ and $y = 2j$. So, $x+y = 2i+2j = 2(i+j)$. Hence, there must exist an integer $k$ such that $x + y = 2k$, namely $k = i + j$. Therefore, $x + y$ is even. $\square$

# Big-Oh

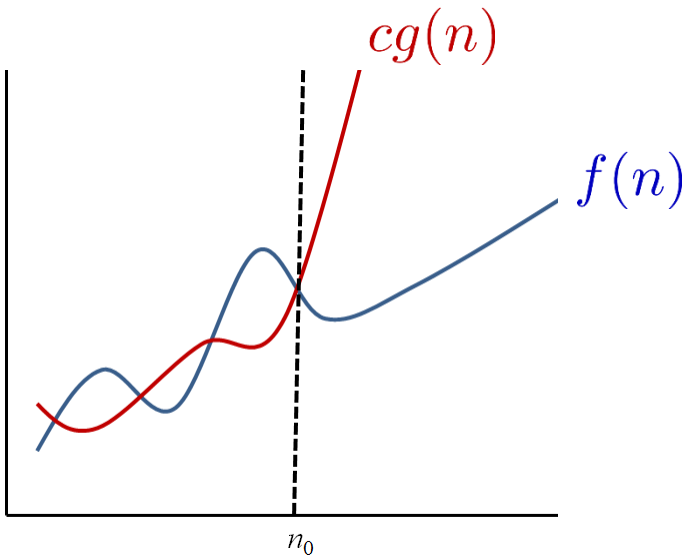
- Upper bound *("at most")*

$f(n) = O(g(n))$ if and only if there exist positive constants $c$ and $n_0$ such that $f(n) \leq cg(n)$ for all $n \geq n_0$.

# Big-Oh in pictures

- Upper bound *("at most")*

$f(n) = O(g(n))$ if and only if there exist positive constants $c$ and $n_0$ such that $f(n) \leq cg(n)$ for all $n \geq n_0$.

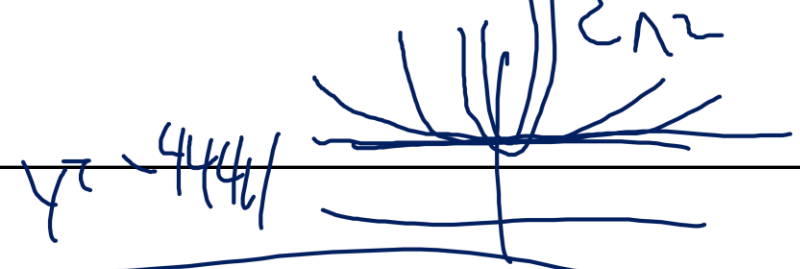



**Translation:** $f$ is smaller than some multiple of $g$ <u>eventually</u> (and <u>stays smaller</u>)

Small values of $n$ don't matter

$f$ isn't growing faster than $g$

# Big-Oh




- Upper bound *("at most")*

$f(n) = O(g(n))$ if and only if there exist positive constants $c$ and $n_0$ such that $f(n) \leq cg(n)$ for all $n \geq n_0$.

- We say "*g(n)* dominates *f(n)*" when *f(n)* = *O(g(n))*
- Notation weirdness:
  - *O*, *Ω*, and *Θ* are classes (sets) of functions
  - BUT: we use = to assign class, not ∈
- **Example**
  - Prove that $7n^2 + 19n - 4444 = O(n^2)$.

7n^2 + 19n - 4444 <= 7n^2 + 19n <= 7n^2 + 19n^2 for all n >= 1
<= 26 * n^2 for all n >= 1

There exist positive constants c and n_0 such that 7n^2 + 19n - 4444 <= c*n^2 for all n >= n_0.
Goal: 7n^2 + 19n - 4444 = O(n^2)

# Big-Oh

- Upper bound *("at most")*

$f(n) = O(g(n))$ if and only if there exist positive constants $c$ and $n_0$ such that $f(n) \leq cg(n)$ for all $n \geq n_0$.

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- **Example**
  - Prove that $7n^2 + 19n - 4444 = O(n^2)$.

*Proof.* If $n \geq 19$,

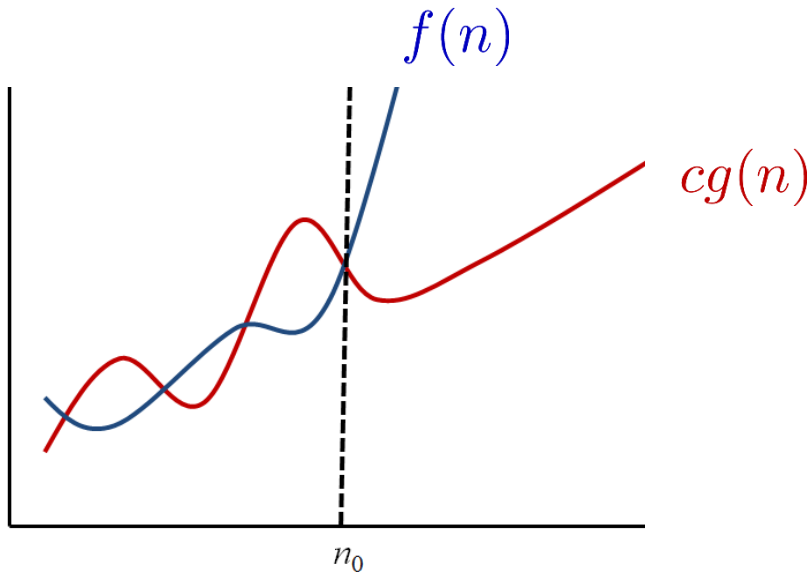
$$\begin{aligned} 7n^2 + 19n - 4444 &\leq 7n^2 + 19n \\ &\leq 7n^2 + n^2 \\ &= 8n^2 \end{aligned}$$

Therefore, there exist positive constants $c = 8$ and $n_0 = 19$ such that $7n^2 + 19n - 4444 \leq cn^2$ for all $n \geq n_0$. □

# Big-Omega picture

- Lower bound *("at least")*

$f(n) = \Omega(g(n))$ if ~~and~~ only if there exist positive constants $c$ and $n_0$ such that $f(n) \geq cg(n)$ for all $n \geq n_0$.

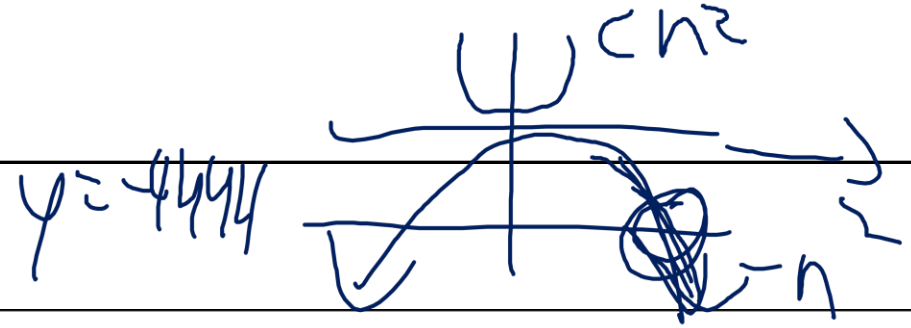



**Translation:** *f* is bigger than some multiple of *g* <u>eventually</u> (and <u>stays bigger</u>)

Small values of *n* don't matter

*g* isn't growing faster than *f*

# Big-Omega




- Lower bound *("at least")*

$f(n) = \Omega(g(n))$ if and only if there exist positive constants $c$ and $n_0$ such that $f(n) \geq cg(n)$ for all $n \geq n_0$.

- **Example**
  - Prove that $7n^2 + 19n - 4444 = \Omega(n^2)$.

7n^2 + 19n - 4444 >= 7n^2 - 4444 for all n >= 0
>= 7n^2 - n^2 for all n >= sqrt(4444)
>= 6n^2 for all n >= sqrt(4444)

There exist positive constants c and n_0 such that:
7n^2 + 19n - 4444 >= c*n^2 for all n >= n_0
Goal:  7n^2 + 19n - 4444 = Omega(n^2)

# Big-Omega

- Lower bound *("at least")*

$f(n) = \Omega(g(n))$ if and only if there exist positive constants $c$ and $n_0$ such that $f(n) \geq cg(n)$ for all $n \geq n_0$.

- **Example**
  - Prove that $7n^2 + 19n - 4444 = \Omega(n^2)$.

*Proof.* If $n \geq 4444$,

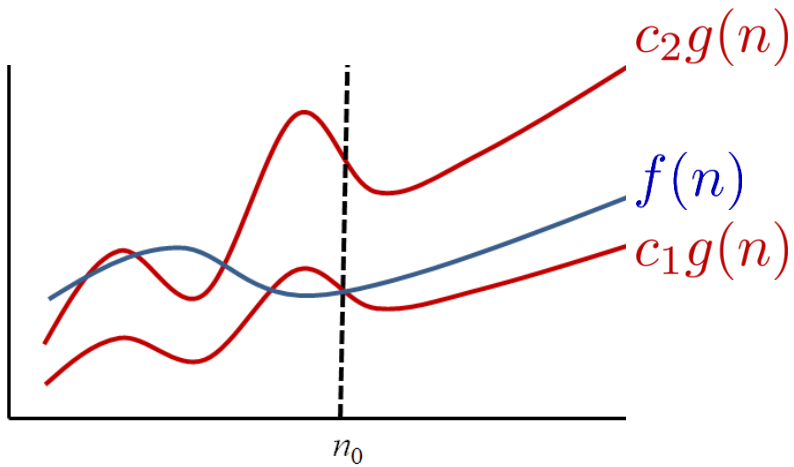
$$\begin{aligned} 7n^2 + 19n - 4444 &\geq 7n^2 - 4444 \\ &\geq 7n^2 - n^2 \\ &= 6n^2 \end{aligned}$$

Therefore, there exist positive constants $c = 6$ and $n_0 = 4444$ such that $7n^2 + 19n - 4444 \geq cn^2$ for all $n \geq n_0$. $\square$

# Big-Theta picture

- Upper *and* lower bound *("same rate as")*

$f(n) = \Theta(g(n))$ if and only if there exist positive constants $c_1$, $c_2$, and $n_0$ such that $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0$.




**Translation:** *f* can be sandwiched between two multiples of *g* <u>eventually</u> (and <u>stays between them</u>)

*f* and *g* are growing at the same rate

Small values of *n* don't matter

# Big-Theta

- Upper *and* lower bound *("same rate as")*

$f(n) = \Theta(g(n))$ if and only if there exist positive constants $c_1$, $c_2$, and $n_0$ such that $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0$.

- **Example**
  - Prove that $7n^2 + 19n - 4444 = \Theta(n^2)$.

6 * n^2 <= 7n^2 + 19n - 4444 for all n >= sqrt(4444)
7n^2 + 19n - 4444 <= 8 * n^2 for all n >= 19
There exist positive constants c1, c2, and n0 such that
c1*n^2 <= 7n^2 + 19n - 4444 <= c2 * n^2 for all n >= n0,
namely c1 = 6, c2 = 8, and n_0 = sqrt(4444).
Goal:  7n^2 + 19n - 4444 = Theta(n^2)

# Big-Theta

- Upper *and* lower bound *("same rate as")*

$f(n) = \Theta(g(n))$ if and only if there exist positive constants $c_1$, $c_2$, and $n_0$ such that $c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0$.

- **Example**
  - Prove that $7n^2 + 19n - 4444 = \Theta(n^2)$.

*Proof.* If $n \geq 4444$,

$$\begin{aligned} 7n^2 + 19n - 4444 &\geq 7n^2 + 19n - n \\ &= 7n^2 + 18n \\ &\geq 7n^2 \\ 7n^2 + 19n - 4444 &\leq 7n^2 + 19n \\ &\leq 7n^2 + n^2 \\ &= 8n^2 \end{aligned}$$

Therefore, there exist positive constants $c_1 = 7$, $c_2 = 8$, and $n_0 = 4444$ such that $c_1 n^2 \leq 7n^2 + 19n - 4444 \leq c_2 n^2$ for all $n \geq n_0$. $\square$

# Analysis of Big-Oh

**Pros**

- Provides a useful summary of the growth rate of the complexity
- Compact
- Simple: eight classes cover most useful algorithms

$$O(1) \ll O(\lg n) \ll O(n) \ll O(n \lg n) \ll O(n^2) \ll O(n^3) \ll O(2^n) \ll O(n!)$$

**Cons**

- Ignores contributions from coefficients and lower-order terms
- Doesn't rank algorithms with same growth rate
- Doesn't rank algorithms on small inputs
- Some of the "best" algorithms have extremely large coefficients, making them impractical for many purposes

# Coming up

- Big-Oh properties
- Algorithm analysis
- Recursive analysis
- Data structures

- **Recommended reading (today):** Sections 1.1 and 1.2
  - *Practice problems:* R-1.3, R-1.7, R-1.19, R-1.21, C-1.8
- **Recommended reading (next week):** Section 1.3
  - *Practice problems:* R-1.11, R-1.15, R-1.26, C-1.3, A-1.4